ENERGY SCIENCES COALITION

Energy Sciences Coalition Response to Request for Information on the Development of an Artificial Intelligence Action Plan

March 11, 2025

The Energy Sciences Coalition (ESC), a broad-based coalition of more than 100 organizations representing the nation's scientific and technical talent at universities, industry, and national laboratories supported by the U.S. Department of Energy (DOE) Office of Science, appreciates the opportunity to provide actionable recommendations to the Office of Science and Technology Policy's February 6 Request for Information to sustain and enhance America's Artificial Intelligence (AI) dominance.

Opportunity

The opportunity provided by AI for addressing the nation's science, energy and national security challenges is vast. Seizing this opportunity will take public-private partnerships between research universities, national labs, federal agencies, and industry that together can drive U.S. leadership. In this context, the DOE Office of Science must play a leading role within a broader federal strategy, and a focused DOE AI Initiative would serve as an essential driver of U.S. AI research and innovation.

DOE already has a clear roadmap through its Frontiers in Artificial Intelligence for Science, Security, and Technology (FASST) initiative to drive AI innovation and make immediate progress in our national interest. The development of this roadmap was influenced by the vision outlined in the 2023 *Advanced Research Directions on AI for Science, Energy, and Security*, dozens of workshops, and Requests for Information with input from a broad set of stakeholders. Examples of key opportunities where AI plays a major role in science and technology solutions include:

- accelerating the search for new materials for quantum computing, sensing, and networking applications;
- developing new nuclear and fusion reactor designs;
- refurbishing nuclear weapons designs to maintain the nation's nuclear deterrent;

- predicting and detecting nuclear proliferation in countries of concern;
- protecting the power grid from cyber attack;
- optimizing energy production of existing energy infrastructure;
- creating predictable ways to engineer microbes and plants for bioproducts; and
- advancing usable computer models and simulations to improve resiliency and mitigate the worst effects of extreme weather events.

DOE's leadership in AI is based on decades of investment in supercomputing and advanced scientific experimental facilities, data management and communications infrastructure, and the scientific workforce.

- **Supercomputing.** DOE builds and operates the world's fastest and most capable supercomputers, including the world's first exascale computers which can perform more than a quintillion tasks per second (a 1 followed by 18 zeroes). These calculations—along with scientific data from other state-of-the-art facilities such as light sources, accelerators, and genomics centers—produce tremendous amounts of data, which are key inputs for AI and machine learning applications across various scientific and national security areas.

- **Data and data infrastructure.** DOE has developed unique data management infrastructure to store, review, and use this data to meet DOE missions. It also stewards key data repositories designed as a public resource to support scientific and technological discoveries for industry, academic, and other federal agencies. with information not just from DOE, but also from industry and other federal agencies. These resources include relevant data repositories for materials, biology and environmental research as well as for residential and commercial buildings, geothermal, solar, wind, marine, and hydrokinetic energy applications.

- **Workforce development.** DOE is also one of the largest federal sponsors of STEM research, education and workforce development, including in applied mathematics and computer science. Investments in these disciplines give young researchers the foundations to understand generative AI and large-scale language models, such as ChatGPT.

Recommendations

ESC recommends establishing a focused, dedicated DOE AI Initiative, aligned with a broader whole-of-government approach to AI research and development, to fully leverage DOE's unique infrastructure and expertise to meet the nation's science, energy, and national security challenges. Key elements should include:

- **Cross-cutting R&D program.** ESC recommends a broad, whole-of-DOE effort that focuses on unique DOE science, engineering, energy, and national security missions. This new program should support fundamental science, early-stage engineering and prototyping of AI hardware and software technologies, and the deployment of AI applications to advance science, energy, and national security. This program should also include flexible and diverse funding mechanisms to drive innovation, mitigate risks, and scale up activities across a range of disciplines and performers, including single principal investigator and small research group grants for high-risk and innovative paths and prototypes; medium-sized efforts to tackle specific grand challenges such as Energy Frontier Research Centers and Scientific Discovery through Advanced Computing (SciDAC) centers; and large-scale consortia of universities, national laboratories, and industry to integrate, test, and deploy AI systems. A key component would be the safe, reliable, and transparent use of AI, including addressing privacy issues. While the program would set its own research directions, it may benefit from broader coordination efforts by the National AI Initiative Coordinating Office within the White House, which can inform gaps, opportunities or partnerships with other federal AI R&D activities.

- **Joint leadership by the Office of Science and National Nuclear Security Administration (NNSA).** ESC recommends that leadership of a DOE AI Initiative be jointly shared by the DOE Office of Science and NNSA to address complementary science and national security missions. The DOE Office of Science should coordinate closely with DOE's applied energy offices to develop energy-specific applications and ensure AI platforms are integrated across DOE missions. The Exascale Computing Initiative serves as a successful model of DOE Office of Science and NNSA collaboration and partnership in designing, building, and operating the first exascale supercomputers. DOE Office of Science and NNSA jointly funded co-design teams of over 1,000 scientists, engineers, and other experts from DOE national laboratories, academia, and industry to work together on hardware, software, and applications. This coordinated framework, coupled with a robust research and deployment effort at scale, is essential to drive innovation in AI.

- **National Science, Energy, and National Security Hubs.** ESC recommends the launch of at least 8 large-scale innovation hubs focused on advancing unique AI applications for DOE science, energy, and national security missions. These large-scale teams of DOE national laboratories, universities, industry, and other research organizations bring together unique DOE research expertise, infrastructure, and workforce to have a significant impact. STEM education and workforce development – as well as fostering connections across a wide range of disciplines ranging from physicists and engineers to computer scientists – would be an important element of this consortia and would complement science and technology research and development efforts. While industry is making significant investments in AI and driving innovation, it is focused on applications central to industry rather than DOE critical mission areas in science, energy, and national security. ESC recommends up to $35 million per year for each Hub in new funding over 10 years to tackle the opportunities and challenges of AI for DOE missions.

- **AI Computing Infrastructure and Integrated Data Management.** ESC recommends significant investments in AI computing infrastructure, such as AI accelerators for hardware and software development. Custom-designed AI hardware and software for DOE computing infrastructure will push the boundaries of technology development and applications and help drive innovation in the private sector. ESC also recommends investments in integrated data management leveraging DOE's vast unclassified and classified data sets to make it available to national lab and academic researchers, industry, state and local governments, communities, and other users to accelerate the pace of innovation and application development and help with decision-making. DOE should also partner with other federal agencies, including the Department of Commerce, to evaluate and mitigate national and global security risks associated with AI systems and the National Science Foundation to coordinate the development of the National AI Research Resource (NAIRR) and other accessible AI data infrastructure.

- **STEM Education and Workforce Development.** ESC recommends boosting targeted investments at DOE in AI STEM education and workforce development. Options include DOE scaling up existing programs, such as the Computational Science Graduate Fellowship Program and the Early Career Research Program or further expanding traineeships to have dedicated tracks for AI to meet the growing gap in an AI-skilled and -trained workforce. As part of this effort, DOE may also look to expand STEM education support for bedrock disciplines like math, physics and chemistry – which are important competencies for a skilled AI workforce – or partner with other agencies like NSF with strong existing STEM initiatives of their own, to expand opportunities for students, postdocs and early-career researchers and serve as a "force multiplier" for federal investments.

- **Authorized funding levels to maintain U.S. competitiveness**. ESC recommends at least $1 billion a year in new funding for a DOE AI Initiative that includes the program elements highlighted above, including investments at the DOE Office of Science, NNSA, and the applied energy offices.

Thank you for your consideration of this critically important initiative.

Sincerely,

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ESC Membership

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